

CONCEPT QUESTIONS – CH : 4

Lumped System Analysis

4-1C

What is the physical significance of the Biot number? Is the Biot number more likely to be larger for highly conducting solids or poorly conducting ones?

The Biot number (Bi) represents the ratio of internal conduction within a solid to the external convection resistance at its surface.

$$Bi = \frac{hL_c}{k}$$

The Biot number is larger for poorly conduction solids since k is small which will increase thermal resistance.

4-2C

What is lumped system analysis? When is it applicable?

The lumped system analysis is a transient heat transfer method that assumes the temperature in a solid is uniform at any given time. It is applicable to when $Bi \leq 0.1$, since the internal conduction resistance is smaller than convection resistance. Heat can redistribute inside the solid faster than it leaves the surface.

4-3C

In what medium is the lumped system analysis more likely to be applicable: in water or in air? Why?

Lumped system analysis is more likely in air since air has a lower convection heat transfer coefficient h . A smaller h results in a smaller Bi number which makes it more likely to satisfy $Bi \leq 0.1$.

4-4C

For which solid is the lumped system analysis more likely to be applicable: an actual apple or a golden apple of the same size? Why?

Lumped system analysis is more applicable to the golden apple since the thermal conductivity for gold is higher than a real apple. A higher thermal conductivity results in a smaller Bi number which makes it more likely to satisfy $Bi \leq 0.1$.

4-5C

For which kind of bodies made of the same material is the lumped system analysis more likely to be applicable: slender ones or well-rounded ones of the same volume? Why?

Lumped system analysis is more likely to be applicable for slender body since a slender body will have a greater surface area which lowers the characteristic length. A smaller characteristic length results in a smaller Bi number which makes it more likely to satisfy $Bi \leq 0.1$.

4-6C

Consider heat transfer between two identical hot solid bodies and the air surrounding them. The first solid is being cooled by a fan while the second one is allowed to cool naturally. For which solid is the lumped system analysis more likely to be applicable? Why?

Lumped system analysis is more likely to be applicable for the solid that is allowed to cool naturally since convection heat transfer coefficient will be smaller. A smaller h results in a smaller Bi number which makes it more likely to satisfy $Bi \leq 0.1$.

4-7C

Consider heat transfer between two identical hot solid bodies and their environments. The first solid is dropped in a large container filled with water, while the second one is allowed to cool naturally in the air. For which solid is the lumped system analysis more likely to be applicable? Why?

Lumped system analysis is more likely to be applicable to the solid cooled naturally in the air since convection heat transfer coefficient for air is smaller compared to water. A smaller h results in a smaller Bi number which makes it more likely to satisfy $Bi \leq 0.1$.

4-8C

Consider a hot baked potato on a plate. The temperature of the potato is observed to drop by 4°C during the first minute. Will the temperature drop during the second minute be less than, equal to, or more than 4°C ? Why?

The temperature drop during the second minute will be less than 4°C since the temperature difference decays exponentially. As the potato cools, the temperature difference $T - T_{\infty}$ becomes smaller.

$$\frac{T(t) - T_{\infty}}{T_i - T_{\infty}} = e^{-bt}$$

4-9C

Consider a potato being baked in an oven that is maintained at a constant temperature. The temperature of the potato is observed to rise by 5°C during the first minute. Will the temperature rise during the second minute be less than, equal to, or more than 5°C ? Why?

The temperature rise during the second minute will be less than 5°C since the temperature difference between the potato and oven decreases over time. As the potato temperature increases the temperature difference $T - T_{\infty}$ becomes smaller.

4-10C

Consider two identical 4kg pieces of roast beef. The first piece is baked as a whole, while the second is baked after being cut into two equal pieces in the same oven. Will there be any difference between the cooking times of the whole and cut roasts? Why?

The roast beef that is cut into two equal pieces will cook faster since the total surface area increases. By increasing the surface area the characteristic length decreases which results in a smaller Bi number. A smaller Bi number allows heat to move through the solid more easily compared to the whole roast beef.

4-11C

Consider a sphere and a cylinder of equal volume made of copper. Both the sphere and the cylinder are initially at the same temperature and are exposed to convection in the same environment. Which do you think will cool faster, the cylinder or the sphere? Why?

The cylinder will cool faster since the surface area of a cylinder is greater than the surface area of a sphere. A greater surface area will result in a smaller characteristic length resulting in a smaller Bi number.

Transient Heat Conduction in Large Plan Walls, Long Cylinders, & Spheres w/ Spatial Effects

4-39C

An egg is to be cooked to a certain level of done-ness by being dropped into boiling water. Can the cooking time be shortened by turning up the heat and bringing water to a more rapid boiling?

No, the cooking time cannot be shortened since at 100°C water reaches its boiling point. Increasing the temperature of boiling water only increases the rate of evaporation not the temperature of the water.

4-40C

What is an infinitely long cylinder? When is it proper to treat an actual cylinder as being infinitely long, and when is it not? For example, is it proper to use this model when finding the temperatures near the bottom or top surfaces of a cylinder? Explain.

An infinitely long cylinder assumes that heat transfer only occurs in the radial direction. It is proper to treat a cylinder as infinitely long when the length is much greater than the radius and the region that is analyzed is far from the ends.

4-41C

What is the physical significance of the Fourier number? Will the Fourier number for a specified heat transfer problem double when the time is doubled?

The Fourier number (F_o) is a dimensionless time that represents the ratio of: Heat Conduction rate within the body to Thermal Energy Storage.

$$F_o = \frac{\alpha t}{L_c^2}$$

Where α is thermal diffusivity, t is time, and L_c is the characteristic Length. The physical significance of Fourier number is that it measures how far heat has diffused relative to the body size. A small F_o means little time has passed and temperature change has not diffused through the body. A large F_o means heat has had time to diffuse through the body.

Yes, the Fourier number will double if time is double since thermal diffusivity, and the characteristic length remain constant, the only variable changing is t . Therefore, the Fourier number must double when time is doubled. Essentially, it means that heat has had twice as much time to spread so the diffusion progression doubles.

4-42C

Why are the transient temperature charts prepared using non-dimensionalized quantities such as the Biot and Fourier numbers instead of the actual variables such as thermal conductivity and time?

Transient charts are dimensionless so that the results are generalized and universal. The Biot and Fourier numbers combine material, size, convection coefficient, and time into 2 dimensionless equations. This allows for the charts to apply any material and geometry as long as the Bi and F_o number are the same. Non-dimensionalization makes solutions universal.

4-44C

How can we use the transient temperature charts when the surface temperature of the geometry is specified instead of the temperature of the surrounding medium and the convection heat transfer coefficient?

The surface temperature is specified when $Bi \Rightarrow \infty$. When Bi is ∞ the convection resistance is effectively zero and the surface temperature is fixed. This causes the boundary conditions to be constant and the temperature condition at the surface to be constant. In order to make Bi go to ∞ the convection heat transfer coefficient is set to ∞ , which is also why the convection resistance goes to zero.

4-45C

The Biot number during a heat transfer process between a sphere and its surroundings is determined to be 0.02. Would you use lumped system analysis or the transient temperature charts when determining the midpoint temperature of the sphere? Why?

Use lumped system analysis since $Bi < 0.1$. A Bi number less than 0.1 means that the temperature within the sphere is approximately uniform. This corresponds to the surface temperature being essentially the same as the midpoint.

4-46C

A body at an initial temperature of T_i is brought into a medium at a constant temperature of T_∞ . How can you determine the maximum possible amount of heat transfer between the body and the surrounding medium?

The maximum amount of heat transfer between the body and the surrounding medium is when the body reaches thermal equilibrium with the surrounding medium.

$$Q_{max} = mc_p(T_i - T_\infty)$$

Where Q_{max} represents the total energy required for the body to cool down or heat up until it reaches T_∞ . When The body reaches the surrounding temperature $T - T_\infty = 0$, therefore heat transfer stops. Maximum heat transfer is equal to the total internal energy to reach equilibrium.

Transient Heat Conduction in Semi-Infinite Solids

4-82C

Under what conditions can a plane wall be treated as a semi-infinite medium?

A plane wall can be treated as a semi-infinite medium when the wall is thick and if we are only interested in the variation of temperature in the region near one of the surfaces. The other side of the wall in a semi-infinite medium would be considered too far away to have any impact on the region of interest during the time of observation.

4-83C

What is a semi-infinite medium? Give examples of solid bodies that can be treated as semi-infinite mediums for heat transfer purposes.

A semi-infinite medium is an idealized body that has a single plane surface and extends to infinity in all directions. The idealized body is used to determine the temperature change in the region we are interested in, close to the surface, due to thermal conditions. Some examples include the ground, a concrete wall/slab, and asphalt pavement. These examples mainly experience heat transfer from one side and is thick enough where the back side of the semi-infinite medium does not experience heat transfer.

4-84C

Consider a hot semi-infinite solid at an initial temperature of T_i that is exposed to convection to a cooler medium at a constant temperature of T_∞ , with a heat transfer coefficient of h . Explain how you can determine the total amount of heat transfer from the solid up to a specified time to.

For a semi-infinite solid, the total amount of heat transfer up to a specified time is determined by how much energy has been lost by the body. Use the error function and the complementary error function to determine the temperature profile using the similarity variable. Knowing the temperature distribution, the heat flux at the surface can be found by Fourier's Law.

$$q_s'' = -k \left. \frac{\partial T}{\partial x} \right|_{x=0} = -k \left. \frac{\partial T}{\partial \eta} \frac{\partial \eta}{\partial x} \right|_{\eta=0} = -k C_1 e^{-\eta^2} \left. \frac{1}{\sqrt{4\alpha t}} \right|_{\eta=0} = \frac{k(T_s - T_i)}{\sqrt{\pi \alpha t}}$$